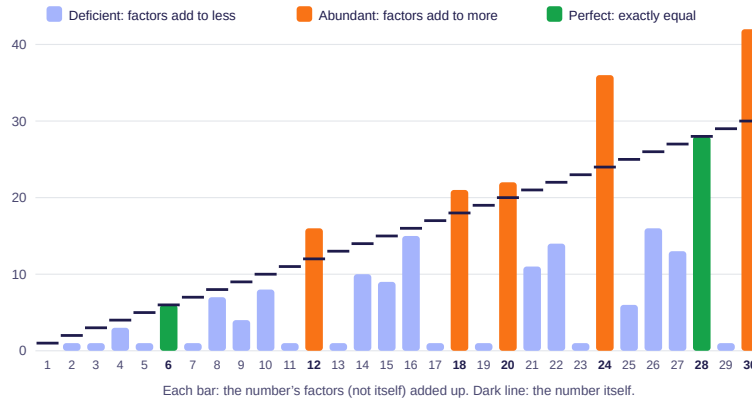


Odd Perfect Numbers

A number is perfect when its factors add up to it, like $6 = 1 + 2 + 3$. Is any perfect number odd?



Bars show each number's factors added up. The dark line is the number itself. Only 6 and 28 match exactly.

TRY IT

6: $1 + 2 + 3 = 6$ (perfect)

28: $1 + 2 + 4 + 7 + 14 = 28$ (perfect)

12: $1 + 2 + 3 + 4 + 6 = 16$ (abundant)

15: $1 + 3 + 5 = 9$ (deficient)

WHAT MATHEMATICIANS KNOW

- Euclid, around 300 BCE, found a recipe that makes even perfect numbers. Leonhard Euler later proved every even perfect number comes from that recipe.
- About fifty perfect numbers are known. Every one is even.
- No odd perfect number exists below 10^{1500} , a 1 followed by 1,500 zeros. Any odd one would need many different prime factors.

WHY IT'S STILL UNSOLVED

Odd numbers usually fall short, since their factors are spread thin, but not always: 945 is the smallest odd number whose factors add to more than itself. So odd numbers can pass their own value, and no one can rule out one landing exactly on it.

TRY THIS WITH STUDENTS

- Sort 1 through 30 into deficient, perfect, and abundant. What do the abundant numbers have in common?
- Ask: Are there odd abundant numbers? (Yes, but the first is 945. A great surprise for students who conjecture "never.")
- Ask: What happens with 8, 16, 32? (Their factors always add to one less.)

WORDS TO KNOW

Factor a number that divides evenly into another

Abundant / deficient factors add to more than / less than the number